JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON THURSDAY 02nd APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

MATHEMATICS — TEST PAPER WITH SOLUTION

SECTION-A

1. Let α, β be the roots of the equation $x^2-3x + r = 0$, and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 + 3x + r = 0$. If the roots of the equation $x^2 + 6x = m$ are $2\alpha + \beta + 2r$ and $\alpha - 2\beta - \frac{r}{2}$, then m is equal to :-

(1) –135 (2) –567

(3) 135 (4) 567

Ans. (4)

Sol. $\alpha + \beta = 3$

$$\frac{\alpha}{2} + 2\beta = -3$$

On solving, we get $\alpha = 6; \beta = -3$

Product of roots = $\alpha\beta = r \Rightarrow r = -18$

Now for $x^2 + 6x - m = 0$

Product of roots = –m

$$= (2\alpha + \beta + 2r)(\alpha - 2\beta - \frac{r}{2})$$

$\Rightarrow -m = (-27)(21)$

$\Rightarrow m = 567$

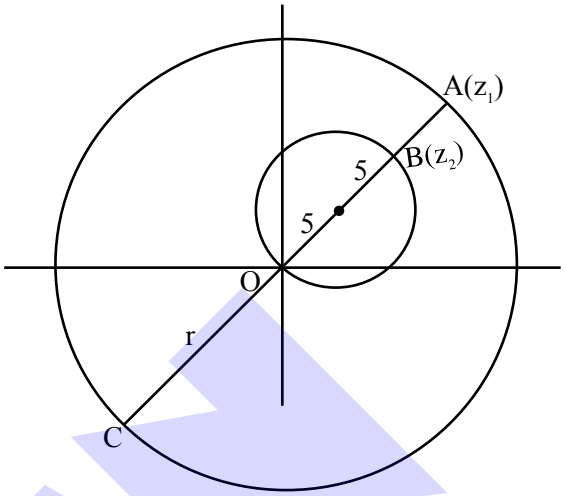
2. Let the circle $C_1 : |z| = r$ and $C_2 : |z - 3 - 4i| = 5$, $z \in \mathbb{C}$, be such that C_2 lies within C_1. If z_1 moves on C_1, z_2 moves on C_2 and min $|z_1 - z_2| = 2$, then max $|z_1 - z_2|$ is equal to :-

(1) 12 (2) 17

(3) 22 (4) 24

Ans. (3)

Sol.



$|z| = r$

$OA = r$

minimum $|z_1 - z_2| = 2$

$\Rightarrow r - 10 = 2 \Rightarrow r = 12$

maximum value of CB

$|z_1 - z_2| = r + 10 = 12 + 10 = 22$

3. If the system of equations

$x + 5y + 6z = 4$,

$2x + 3y + 4z = 7$,

$x + 6y + az = b$

has infinitely many solutions, then the point (a, b) lies on the line

(1) y – x = 3 (2) x – y = 3

(3) x + y = 11 (4) x + y = 12

Ans. (2)

Sol. $D = \begin{vmatrix} 1 & 5 & 6 \\ 2 & 3 & 4 \\ 1 & 6 & a \end{vmatrix} = 0$

$$\Rightarrow a = \frac{50}{7}$$

$$D_z = 0 \Rightarrow \begin{vmatrix} 1 & 5 & 4 \\ 2 & 3 & 7 \\ 1 & 6 & b \end{vmatrix} = 0$$

$$\Rightarrow b = \frac{29}{7}$$

$\therefore$ (a, b) lies on x – y = 3

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4. Let a_1,a_2,a_3, ... be an A.P. and $g_1 = a_1,g_2,g_3$, ... be an increasing G.P. If $a_1 = a_2 + g_2 = 1$ and $a_3 + g_3 = 4$, then $a_{10} + g_5$ is equal to :

(1) 81 (2) 76

(3) 62 (4) 55

Ans. **(4)**

Sol. A.P. : $1, a_2, a_3, \ldots$

G.P. : $1, g_2, g_3, \ldots$

$a_2 + g_2 = 1 \Rightarrow 1 + d + r = 1 \Rightarrow d + r = 0$

$1 + 2d + r^2 = 4 \Rightarrow r^2 - 2r - 3 = 0$

$\Rightarrow r = 3, -1$ (reject)

$\therefore d = -3$

$\therefore a_{10} = 1 + 9d = 1 + 9(-3) = -26$

$g_5 = 1.r^4 = 3^4 = 81$

$\therefore a_{10} + g_5 = 55$

5. The sum $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \ldots$ up to 8 terms, is :

(1) 70 (2) 71

(3) 72 (4) 73

Ans. **(2)**

Sol. $T_r = \frac{1^3+2^3+3^3+\ldots..r^3}{1+3+5+\ldots..+(2r-1)} = \frac{\left(\frac{r(r+1)}{2}\right)^2}{r^2}$

$= \frac{r^2+2r+1}{4}$

$S_n = \sum_{r=1}^{n} T_r$

$S_n = \frac{1}{4}\sum\left(r^2+2r+1\right)$

$= \frac{1}{4}\left[\frac{n(n+1)(2n+1)}{6} + 2\frac{n(n+1)}{2} + n\right]$

$S_8 = \frac{1}{4}\left[\frac{8\times9\times17}{6} + 8\times9 + 8\right]$

$= \frac{1}{4}[204 + 72 + 8] = 71$

6. If for $3 \le r \le 30$,

$({}^{30}C_{30-r})+3({}^{30}C_{31-r})+3({}^{30}C_{32-r})+({}^{30}C_{33-r}) = {}^{m}C_r$,

then m equals :

(1) 31 (2) 32

(3) 33 (4) 34

Ans. **(3)**

Sol. ${}^3C_3 \cdot {}^{30}C_{30-r} + {}^3C_2 \cdot {}^{30}C_{31-r} + {}^3C_1 \cdot {}^{30}C_{32-r} + {}^3C_0 \cdot {}^{30}C_{33-r}$

$= {}^{33}C_{33-r} \equiv {}^{m}C_r \equiv {}^{m}C_{m-r}$

$\Rightarrow m = 33$

7. Let p_n denote the total number of triangles formed by joining the vertices of an n-side regular polygon. If $p_{n+1} - p_n = 66$, then the sum of all distinct prime divisors of n is :

(1) 7 (2) 8

(3) 5 (4) 6

Ans. **(3)**

Sol. $P_n = {}^nC_3$

$P_{n+1} - P_n = 66$

${}^{n+1}C_3 - {}^nC_3 = 66$

$\Rightarrow \frac{(n+1)n(n-1)}{6} - \frac{n(n-1)(n-2)}{6} = 66$

$\Rightarrow \frac{n(n-1)}{6}[n+1-n+2] = 66$

$\Rightarrow n(n-1) = 132$

$n = 12$

Prime divisors of 12 are 2 & 3

sum = 2 + 3 = 5

8. A man throws a fair coin repeatedly. He gets 10 points for each head he throws and 5 points for each tail he throws. If the probability that he gets exactly 30 points is $\frac{m}{n}$, gcd (m, n) = 1, then m + n is equal to :

(1) 53 (2) 55

(3) 107 (4) 105

Ans. **(3)**

Sol. P = (6tail)+(4tail+1head) + (2tail+2head) + (3head)

$P = \left(\frac{1}{2}\right)^6 + \frac{5!}{1!4!}\left(\frac{1}{2}\right)^5 + \frac{4!}{2!2!}\left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^3$

$P = \frac{1}{64} + \frac{5}{32} + \frac{3}{8} + \frac{1}{8}$

$P = \frac{1+10+24+8}{64} = \frac{43}{64} = \frac{m}{n}$

m + n = 107

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9. The mean and variance of n observations are 8 and 16, respectively. If the sum of the first (n – 1) observations is 48 and the sum of squares of the first (n – 1) observations is 496, then the value of n is :

(1) 21 (2) 16

(3) 13 (4) 7

Ans. (4)

Sol. $x_1 + x_2 + \ldots + x_{n-1} + x_n = 8n$

$48 + x_n = 8n$

$\Rightarrow x_n = 8n - 48 \quad \ldots(1)$

$\Rightarrow 16 = \frac{496 + x_n^2}{n} - (8)^2$

$\Rightarrow 80n = 496 + x_n^2$

$\Rightarrow x_n^2 = 80n - 496 \quad \ldots(2)$

$\Rightarrow (8n - 48)^2 = 80n - 496$

$\Rightarrow 64(n - 6)^2 = 8(10n - 62)$

$\Rightarrow 8(n - 6)^2 = 10n - 62$

$\Rightarrow 4(n - 6)^2 = 5n - 31$

$\Rightarrow 4(n^2 - 36 - 12n) = 5n - 31$

$\Rightarrow 4n^2 + 144 - 48n = 5n - 31$

$\Rightarrow 4n^2 - 53n + 175 = 0$

$\Rightarrow 4n^2 - 28n - 25n + 175 = 0$

$\Rightarrow 4n(n - 7) - 25(n - 7) = 0$

$n = 7$

10. Let a circle pass through the origin and its centre be the point of intersection of two mutually perpendicular lines $x + (k - 1)y + 3 = 0$ and $2x + k^2y - 4 = 0$. If the line $x - y + 2 = 0$ intersects the circle at the points A and B, then $(AB)^2$ is equal to :

(1) 10 (2) 27

(3) 18 (4) 34

Ans. (3)

Sol. $x + (k - 1)y + 3 = 0$

$2x + k^2y - 4 = 0$

$\left(\frac{1}{1-k}\right)\left(\frac{2}{k^2}\right) = 1$

$2 = k^2 - k^3$

$k^3 - k^2 + 2 = 0$

$k = -1$

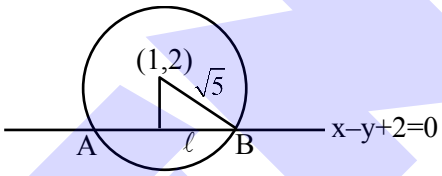
Solving :

$2(x - 2y + 3) = 0$

$2x + y - 4 = 0$

$-5y + 10 = 0$

$y = 2$



$x = 1$

Centre (1, 2)

$r = \sqrt{5}$

So circle is $(x - 1)^2 + (y - 2)^2 = 5$

Chord $x - y + 2 = 0$

$p = \frac{1}{\sqrt{2}}$

$\ell = \sqrt{5 - \frac{1}{2}} = \frac{3}{\sqrt{2}}$

$AB = \frac{6}{\sqrt{2}} \quad \therefore AB^2 = 18$

11. Let O be the origin, and P and Q be two points on the rectangular hyperbola $xy = 12$ such that the mid point of the line segment PQ is $\left(\frac{1}{2}, -\frac{1}{2}\right)$. Then the area of the triangle OPQ equals :

(1) $\frac{3}{2}$ (2) $\frac{5}{2}$

(3) $\frac{7}{2}$ (4) $\frac{9}{2}$

Ans. (3)

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Sol. Equation of chord whose midpoint is $\left(\frac{1}{2},\frac{-1}{2}\right)$

$$\frac{x}{x_1}+\frac{y}{y_1}=2$$

$2x - 2y = 2$

$x - y = 1$

solving with $xy = 12$

$P(-3, -4)$, $Q(4, 3)$

$$\text{Area } (\Delta OPQ) = \frac{1}{2}|-3\times 3-(-4)(4)| = \frac{7}{2}$$

12. Let the parabola $y = x^2 + px + q$ passing through the point $(1, -1)$ be such that the distance between its vertex and the x-axis is minimum. Then the value of $p^2 + q^2$ is :

(1) 2 (2) 4

(3) 5 (4) 8

Ans. (2)

Sol. parabola passes through $(1, -1)$ so

$-1 = 1 + p + q$

$p + q = -2$

$q = -2 - p$ …(1)

Distance from x-axis

$$\frac{-D}{4a}=\frac{-(p^2-4q)}{4(1)}=\frac{4q-p^2}{4}$$

$$=\frac{4(-2-p)-p^2}{4}$$ from (1)

$$=\frac{-8-4p-p^2}{4}=\frac{-(p^2+4p+8)}{4}=\frac{-((p+2)^2+4)}{4}$$

Minimum at $p = -2 \Rightarrow q = 0$

$p^2 + q^2 = 4 + 0 = 4$

13. Let $P = \{\theta \in [0, 4\pi] : \tan^2\theta \neq 1\}$ and $S = \{a \in Z : 2(\cos^8\theta - \sin^8\theta)\sec 2\theta = a^2, \theta \in P\}$. Then n(S) is :

(1) 0 (2) 1

(3) 2 (4) 4

Ans. (1)

Sol. $2(\cos^8\theta - \sin^8\theta)\sec 2\theta = a^2$

$2(\cos^4\theta+\sin^4\theta)(\cos^2\theta+\sin^2\theta)(\cos^2\theta-\sin^2\theta)\sec 2\theta = a^2$

$2(\cos^4\theta + \sin^4\theta) = a^2$

$2(1 - 2\sin^2\theta\cos^2\theta) = a^2$

$$2\left(1-\frac{\sin^2 2\theta}{2}\right)=a^2,\ a\in Z$$

$a^2 = 2 - \sin^2 2\theta \in [1, 2]$

$a^2 = 1$ at $\sin^2 2\theta = 1$

$$2\theta = (2n+1)\frac{\pi}{2}$$

$$\theta = (2n+1)\frac{\pi}{4} \notin P$$

$n(S) = 0$

14. Let the vectors $\vec{a}=-\hat{i}+\hat{j}+3\hat{k}$ and $\vec{b}=\hat{i}+3\hat{j}+\hat{k}$. For some $\lambda,\mu\in\mathbf{R}$, let $\vec{c}=\lambda\vec{a}+\mu\vec{b}$. If $\vec{c}.\left(3\hat{i}-6\hat{j}+2\hat{k}\right)=10$ and $\vec{c}.\left(\hat{i}+\hat{j}+\hat{k}\right)=-2$, then $|\vec{c}|^2$ is equal to :

(1) 8 (2) 12

(3) 14 (4) 15

Ans. (2)

Sol. $\vec{c}=\lambda\vec{a}+\mu\vec{b}$

$$\vec{c}=(\mu-\lambda)\hat{i}+(\lambda+3\mu)\hat{j}+(3\lambda+\mu)\hat{k}$$

$$\vec{c}.(3\hat{i}-6\hat{j}+2\hat{k})=10$$

$3(\mu - \lambda) - 6(\lambda + 3\mu) + 2(3\lambda + \mu) = 10$

$-3\lambda - 13\mu = 10$ …(1)

$$\vec{c}.(\hat{i}+\hat{j}+\hat{k})=-2$$

$5\mu + 3\lambda = -2$ …(2)

Solving (1) & (2)

$\lambda = 1, \mu = -1$

$$\vec{c}=-2\hat{i}-2\hat{j}+2\hat{k}$$

$$|\vec{c}|^2=12$$

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15. Let the point A be the foot of perpendicular drawn from the point P(a, b, 0) on the line $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-\alpha}{3}$. If the midpoint of the line segment PA is $\left(0,\frac{3}{4},\frac{-1}{4}\right)$, then the value of $a^2+b^2+\alpha^2$ is equal to :

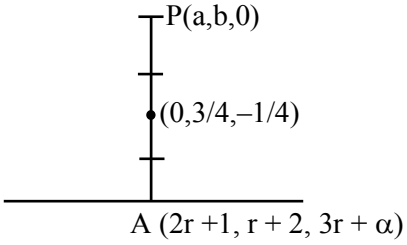
(1) 1 (2) 2

(3) 6 (4) 9

Ans. (1)

Sol. $\frac{2r+1+a}{2}=0 \Rightarrow 2r+a=-1$

$\frac{r+2+b}{2}=\frac{3}{4} \Rightarrow r+b=-\frac{1}{2}$



$\frac{3r+\alpha}{2}=\frac{-1}{4} \Rightarrow 3r+\alpha=\frac{-1}{2}$

$2\cdot a+1\cdot(b-3/4)+3\cdot\frac{1}{4}=0 \Rightarrow a+b=0$

$\Rightarrow 2(-1-2r)+(-r-1/2)=0$

$\Rightarrow -3/2=3r=1 \Rightarrow r=-1/2,\ a=0,\ b=0,\ \alpha=1$

$\therefore a^2+b^2+\alpha^2=1$

16. Two adjacent sides of a parallelogram PQRS are given by $\overrightarrow{PQ}=\hat{j}+\hat{k}$ and $\overrightarrow{PS}=\hat{i}-\hat{j}$. If the side PS is rotated about the point P by an acute angle α in the plane of the parallelogram so that it becomes perpendicular to the side PQ, then $\sin^2\left(\frac{5\alpha}{2}\right)-\sin^2\left(\frac{\alpha}{2}\right)$ is equal to :

(1) $\frac{1}{2}$ (2) $\frac{\sqrt{3}}{2}$

(3) $\frac{\sqrt{3}}{4}$ (4) $\frac{2\sqrt{3}}{5}$

Ans. (2)

Sol. Let angle between $\overrightarrow{PQ}$ & $\overrightarrow{PS}$ is θ

$\cos\theta=\frac{\overrightarrow{PQ}.\overrightarrow{PS}}{\left|\overrightarrow{PQ}\right|\left|\overrightarrow{PS}\right|}$

$\cos\theta=\frac{0-1+0}{\sqrt{2}\sqrt{2}}$

$\cos\theta=\frac{-1}{2}$

$\theta=\frac{2\pi}{3}$

$\alpha=\frac{2\pi}{3}-\frac{\pi}{2}$ So $\alpha=\frac{\pi}{6}$

$\sin^2\frac{5\alpha}{2}-\sin^2\frac{\alpha}{2}=\sin\left(\frac{5\alpha}{2}+\frac{\alpha}{2}\right)\sin\left(\frac{5\alpha}{2}-\frac{\alpha}{2}\right)$

$=\sin\frac{6\alpha}{2}\times\sin\frac{4\alpha}{2}$

$=\sin3\alpha\times\sin2\alpha$

$=\sin\frac{\pi}{2}\times\sin\frac{\pi}{3}$

$=1\times\frac{\sqrt{3}}{2}=\frac{\sqrt{3}}{2}$

17. The value of $\int_0^{20\pi}(\sin^4x+\cos^4x)dx$ is equal to:

(1) $\frac{15\pi}{2}$ (2) 25π

(3) 15π (4) $\frac{25\pi}{2}$

Ans. (3)

Sol. $I=\int_0^{20\pi}(1-2\sin^2x\cos^2x)\cdot dx$

$=20\pi-\frac{1}{2}\int_0^{20\pi}\sin^2 2x\,dx$

$20\pi-\frac{1}{2}(40)\int_0^{\frac{\pi}{2}}\sin^2 2x$

$20\pi-\frac{20}{2}\left(x-\frac{\sin2x}{2}\right)_0^{\frac{\pi}{2}}$

$20\pi-10\left(\frac{\pi}{2}\right)=15\pi$

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18. Let f(x) be a polynomial of degree 5, and have extrema at x = 1 and x = –1. If $\lim_{x\to0}\left(\frac{f(x)}{x^3}\right)=-5$, then f(2) – f(–2) is equal to :

(1) 0 (2) 50

(3) 92 (4) 112

Ans. (4)

Sol. Given f(x) is a polynomial of degree 5

Also, f′(1) = 0; f′(–1) = 0.

Also $\because \lim_{x\to0}\frac{f(x)}{x^3}=-5$ (fixed and finite)

$\therefore$ f(0) = 0; f′(0) = 0; f″(0) = 0

$\frac{f'''(0)}{6}=-5 \Rightarrow f''(0)=-30$

Hence, let f′(x) = (ax + b)(x – 0)(x – 1)(x + 1)

$\Rightarrow f'(x)=ax^4+bx^3-ax^2-bx$

$f''(x)=4ax^3+3bx^2-2ax-b$

$\because f''(0)=0 \Rightarrow b=0$

$f''(x)=12ax^2+6bx-2a$

$\because f''(0)=-2a=-30$

$\Rightarrow a=15$

$\therefore f'(x)=15x.x.(x-1)(x+1)$

$\Rightarrow f'(x)=15x^4-15x^2$

$\Rightarrow f(x)=3x^5-5x^3+C$

$\because f(0)=0 \Rightarrow C=0$

$\therefore f(x)=3x^5-5x^3$

$\therefore f(2)-f(-2)=112$

19. Let $f(x)=\int\left(\frac{16x+24}{x^2+2x-15}\right)dx$. If f(4) = $14\log_e(3)$ and f(7) = $\log_e(2^\alpha.3^\beta)$, α, β ∈ **N**, then α + β is equal to:

(1) 31 (2) 37

(3) 39 (4) 41

Ans. (3)

Sol. $f(x)=\int\frac{8(2x+2)+8}{x^2+2x-15}dx$

$\Rightarrow f(x)=8\ell n|x^2+2x-15|+\ell n\left|\frac{x-3}{x+5}\right|+C$

$\Rightarrow f(4)=14\ell n3+C$

$\Rightarrow 14\ell n3=14\ell n3+C$

$\therefore C=0$

Now, f(7) = 8ℓn48 – ℓn3

$=\ell n\left(\frac{(48)^8}{3}\right)$

$\Rightarrow f(7)=\ell n(2^{32}\times3^7)$

α = 32, β = 7 ⇒ α + β = 39

20. Let x = x(y) be the solution of the differential equation $2y^2\frac{dx}{dy}-2xy+x^2=0$, y > 1 , x(e) = e. Then x(e²) is equal to:

(1) $\frac{3}{2}e^2$ (2) $\frac{2}{3}e^2$

(3) e^2 (4) $2e^2$

Ans. (2)

Sol. 2y(ydx – xdy) + x²dy = 0

$\Rightarrow -2yx^2\ d\left(\frac{y}{x}\right)+x^2dy=0$

$\Rightarrow -2y\ d\left(\frac{y}{x}\right)+dy=0$

$\Rightarrow -2d\left(\frac{y}{x}\right)+\frac{1}{y}dy=0$

$\Rightarrow \frac{-2y}{x}+\log_e y=C$

Given x(e) = e

$\Rightarrow C=-1$

$\therefore \frac{-2y}{x}+\log_e y=-1$

$\Rightarrow \frac{2y}{x}-\log_e y=1$

Put $y=e^2$, we get

$\Rightarrow \frac{2e^2}{x}-2=1 \Rightarrow \frac{2e^2}{x}=3 \Rightarrow x=\frac{2e^2}{3}$

$\Rightarrow x(e^2)=\frac{2e^2}{3}$

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SECTION-B

21. Let A = {2, 3, 4, 5, 6}. Let R be a relation on the set A × A given by (x, y)R(z, w) if and only if x divides z and y ≤ w. Then the number of elements in R is _____.

Ans. **(120)**

Sol. For x = 2, z = 2, 4, 6

x = 3 z = 3, 6

x = 4 z = 4

x = 5 z = 5

x = 6 z = 6

Total case = 8

and no. of combination of y and w satisfying y ≤ w

= 1 +2 + 3 + 4 + 5 = 15

No. of relations satisfying (x,y) R(z,w) is 8 × 15 = 120.

22. Consider the matrices $A = \begin{bmatrix} 2 & -2 \\ 4 & -2 \end{bmatrix}$ and $B\begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}$. If matrices P and Q are such that PA = B and AQ = B, then the absolute value of the sum of the diagonal elements of 2(P + Q) is _____.

Ans. **(34)**

Sol. $A^{-1} = \frac{1}{4}\begin{bmatrix} -2 & 2 \\ -4 & 2 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}$

$\because PA = B$

$\Rightarrow P = BA^{-1}$

$= \frac{1}{2}\begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -21 & 12 \\ -7 & 4 \end{bmatrix}$

$\because AQ = B$

$\Rightarrow Q = A^{-1}B$

$= \frac{1}{2}\begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -2 & -6 \\ -5 & -15 \end{bmatrix}$

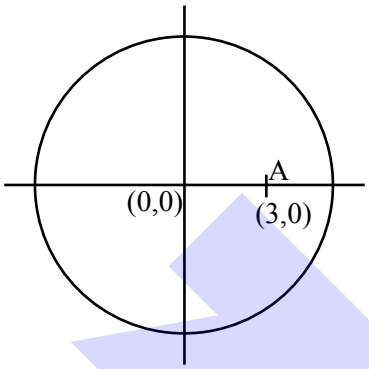
$\therefore |\text{tr}(2(P + Q))| = |(-21) + (4) + (-2) + (-15)|$

$= |-34| = 34$

23. Let A be the point (3, 0) and circles with variable diameter AB touch the circle $x^2 + y^2 = 36$ internally. Let the curve C be the locus of the point B. If the eccentricity of C is e, then $72e^2$ is equal to ____.

Ans. **(18)**

Sol.



Let B(h, k)

Equation of circle with AB as diameter

$(x - h)(x - 3) + (y - k)(y - 0) = 0$

$x^2 + y^2 - (h + 3)x - ky + 3h = 0$

Centre $\left(\frac{h+3}{2}, \frac{k}{2}\right)$

Circle touches internally

$\therefore C_1C_2 = |R - r|$

$$\sqrt{\left(\frac{h+3}{2}\right)^2 + \left(\frac{k}{2}\right)^2} = \left|6 - \frac{1}{2}\left(\sqrt{(h-3)^2 + k^2}\right)\right|$$

$$\sqrt{(x+3)^2 + y^2} + \sqrt{(x-3)^2 + y^2} = 12$$

2a = 12 $\therefore$ a = 6

(–3, 0) and (3, 0) are foci

2ae = 6

12e = 6

$e = \frac{1}{2}$

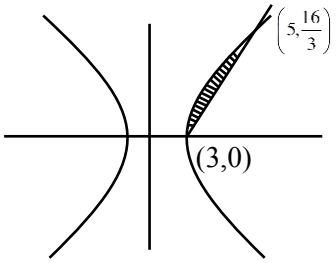
$\therefore 72e^2 = 72\left(\frac{1}{4}\right) = 18$

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24. If the area of the region bounded by $16x^2 - 9y^2 = 144$ and $8x - 3y = 24$ is A, then $3(A + 6\log_e(3))$ is equal to_____.

Ans. **(24)**

Sol.



$16x^2 - 9y^2 = 144$

$16x^2 - (8x - 24)^2 = 144$

$16x^2 - 64(x - 3)^2 = 144 \Rightarrow x^2 - 4(x - 3)^2 = 9$

$3x^2 - 24x + 45 \Rightarrow x^2 - 8x + 15 = 0$

$x = 3,5$

$$\text{Area} = \int_3^5 \sqrt{\frac{16x^2 - 144}{9}} - \frac{1}{2}\cdot 2\cdot\frac{16}{3}$$

$$= \frac{4}{3}\int_3^5 \sqrt{x^2 - 9} - \frac{16}{3}$$

$$= \frac{4}{3}\left(\frac{x}{2}\sqrt{x^2 - 9} - \frac{9}{2}\log_e\left(x + \sqrt{x^2 - 9}\right)\right)_3^5 - \frac{16}{3}$$

$$= \frac{4}{3}\left(\frac{5}{2}\cdot 4 - \frac{9}{2}\log_e 9 - \frac{3}{2}\cdot 0 + \frac{9}{2}\log_e 3\right) - \frac{16}{3}$$

Area $= 8 - 6\log_e 3 = A$

$\therefore A + 6\ell n3 = 8$

$\Rightarrow 3(A + 6\ell n3) = 24$

25. The number of points in the interval [2, 4], at which the function $f(x) = \left[x^2 - x - \frac{1}{2}\right]$, where $[\cdot]$ denotes the greatest integer function, is discontinuous, is_____.

Ans. **(10)**

Sol. In $x \in [2, 4]$, range of $x^2 - x - \frac{1}{2}$ is [1.5, 11.5]

f(x) is discontinuous at points where $x^2 - x - \frac{1}{2}$ becomes integers.

$\therefore$ Integral values of $x^2 - x - \frac{1}{2}$ are 2, 3, 4, 5, 6, 7, 8, 9 10, 11

$\therefore$ Number of points where is discontinuous is 10

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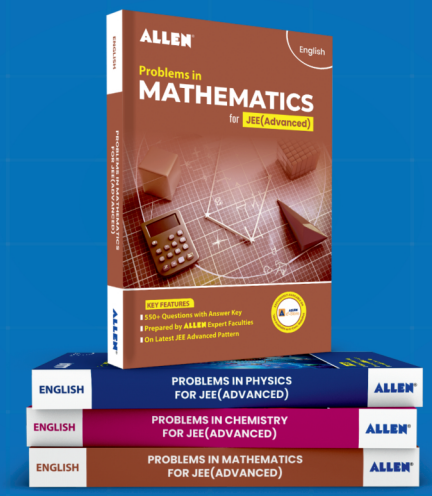


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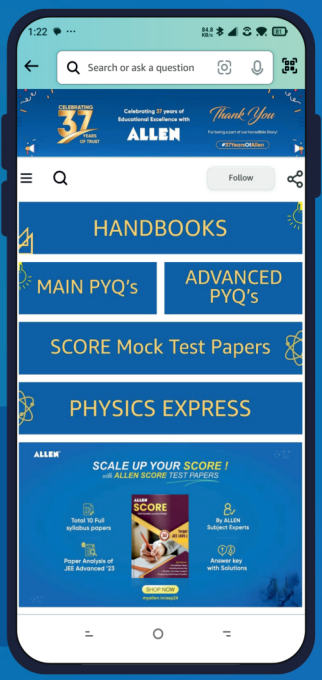
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